

SAI AKILESH VENIGALLA

New York, NY 10012 | 4+ Years of SWE Experience

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Experience

Software Engineer

Jan 2026 - Present

Tesla | *Building mission-critical products and features for the RoboTaxi and Supercharging teams.*

- Launched Robotaxi **shuttle** feature, eliminating **700K daily heap allocations** via H3 indexing, pre-computed geofences, and **Redis** caching for real-time validation to maintain **ultra-low latency** on the **ride-monetization** critical path.
- Achieved **80% faster ride data retrieval** via a high-throughput **MySQL** to **Elasticsearch CDC** sync pipeline.
- Architected a generic **Go, Kafka** pipeline for **asynchronous event processing**, decoupling auxiliary tasks with a Dead Letter Queue (**DLQ**) for **fault-tolerant retries** to guarantee 100% core Robotaxi billing reliability.
- Scaled referral device fingerprinting to **3.5M+ DAU** using **Redis**, idempotency and a secure Apple Pay pre-auth engine.
- Optimized routes for **8M+ users** by pushing real-time Supercharger wait forecasts to **Google Maps** via **OCPI, gRPC**.
- Enabled seamless charging for **350K+ users** in Europe across **1,500+ CPOs** by building a resilient **Multipass** and pricing sync pipeline using **DLQs** and **exponential backoff**.

Software Engineer 2

Sept 2023 - Aug 2024

Citrix | *Led new feature development and backend optimizations for Unified Security.*

- Optimized API response times** by over **95%** (4min to 9s) by re-architecting cloud backend services in **C++**.
- Drove a **60% increase** in application security license sales by architecting a security recommendation microservice in **Python** and **C++**, deployed on **Docker** and **Kubernetes**, while also serving as the team's **Scrum Master**.
- Saved the threat research team 30+ hours of manual work and **minimized CPU usage by 30%** by leading the **integration of LLMs** (OpenAI) with **Python** and **C** to automate WAF signature extraction and categorization.
- Resolved database bottlenecks with **Splunk**, strategic **indexing** and **refactoring**, enhancing backend efficiency by 66%.
- Mentored and Trained 5 software engineers in Agile, OOP, System Design, and conducted detailed Code Walkthroughs.

Software Engineer 1

Jul 2021 - Sept 2023

Citrix | *Drove innovation by architecting large-scale parallel and distributed systems for application security.*

- Slashed** Software Development Life Cycle **runtime by 88%** (16 hrs to under 2 hrs) by architecting a large-scale CI/CD pipeline using Jenkins, **Java**, **Python**, **Groovy** and **Bash Scripting**, earning the **Award** for Technical Innovation.
- Boosted build cycles **30x (1/wk to 30/wk)** with an adaptive scheduling module that parallelized distributed workloads.
- Eliminated 100%** of false negatives with a context-aware Java module that adaptively selects Ruby and Perl test cases.
- Accelerated** container **recovery time by 75%** by using **Prometheus** and **Grafana** for monitoring to trigger **Kubernetes** orchestrated disaster recovery actions.

Software Engineering Intern

Jan 2021 - Jul 2021

Citrix | *Developed new features and built full-stack solutions to accelerate Netscaler's development lifecycle.*

- Cut Netscaler kernel test time by 96%** by building a full-stack web app (**Flask**, **React**) for rapid isolated testing.
- Reduced server load by 20% by blocking malicious bots with browser fingerprinting in **JavaScript** and **C++**.

Projects

WeaveAI | *ReactJS, FastAPI, LangChain, LLMs, MongoDB, ChromaDB, Redis, RAG, Prompt Engineering, GenAI* [Link](#)

- Built an AI learning platform (ReactJS, FastAPI) using LangChain to orchestrate LLMs for custom curricula generation.
- Engineered a **RAG pipeline** with a multi-database backend (ChromaDB, MongoDB, Redis) and automated podcast/quiz generation, **reducing tutor response time by 95%** and boosting user engagement by 40%.

VibeTrader | *ReactJS, Flask, Kafka, SparkNLP, LSTM, TensorFlow, MLOps, Spark, AI Agents, Dask, Finance* [Link](#)

- Developed a smart trading platform fusing market data with a tri-modal sentiment engine to improve stock predictions.
- Implemented an **end-to-end ETL pipeline** using Selenium & PySpark and a Kafka streaming architecture on a multi-database backend (PostgreSQL, MongoDB, Redis) for live alerts & improving inference optimization by 70%.

Technical Skills

- Languages:** Go, Python, Java, C, C++, C#, HTML/CSS, JavaScript, SQL, Typescript, Ruby, Bash/Shell Scripting
- Databases:** PostgreSQL, MongoDB (NoSQL), Amazon S3, Pinecone (Vector Databases), Redis, DynamoDB
- Technologies/Frameworks:** Docker, Kubernetes, Jenkins CI/CD, AWS, Azure, GCP, FastAPI, Flask, Django, ReactJS, Git (Version Control), JIRA, TensorFlow, PyTorch, Spark, Kafka, Elasticsearch, Logstash, Kibana, Terraform, Cursor
- Skills:** Fullstack Development, Backend Development, System Design, Microservices, REST API, Kernel Development

Education

New York University | *Master of Science in Computer Science* | GPA: 3.95/4.0

May 2026

Manipal Institute of Technology | *Bachelor of Science in Computer Science* | GPA: 3.95/4.0

May 2021